

$$\frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx^2 - 3 = 0$$

$$c = 0.1$$

$$k = 3$$

- 1) Define equilibria
- 2) Put in state space form (linearize)
- 3) Draw state space

[20]
[30]
[50]

$$1) \frac{d^2x}{dt^2} + 0.1 \frac{dx}{dt} + 3x^2 - 3 = 0$$

$$x = [\dot{x} \ x]^T \text{ then } \dot{x} = [\ddot{x} \ \dot{x}]^T$$

$$\therefore \dot{x} = R(x)$$

$$\therefore \begin{bmatrix} \ddot{x} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} -0.1\dot{x} - 3x^2 + 3 \\ \dot{x} \end{bmatrix}$$

$$\text{At equilibria, } \dot{x}_0 = R(x_0) = 0$$

$$\therefore \ddot{x} = 0, \dot{x} = 0$$

$$\therefore -3x^2 + 3 = 0 \quad \therefore x = \pm 1$$

2) Apply a small perturbation at equilibria

$$\dot{x}_0 + \dot{x}' = R(x_0 + x')$$

$$\therefore \dot{x}' = x' R_{x_0}$$

$$\therefore \begin{bmatrix} \ddot{x} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} -0.1 & -6 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ x \end{bmatrix}$$

3) Let $x' = x^* e^{\lambda t}$

$$\therefore \dot{x}' = \lambda x' = x' R_{x_0} \quad \therefore (R_{x_0} - I\lambda) x' = 0$$

$$\therefore |R_{x_0} - \lambda I| = 0 \quad \therefore x = x_0 + \sum k_i \xi_i e^{\lambda_i t}$$

for $x = 1$,

$$\begin{bmatrix} \ddot{x} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} -0.1 & -6 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ x \end{bmatrix}$$

$$\therefore \begin{vmatrix} -0.1 - \lambda & -6 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\therefore \lambda^2 + 0.1\lambda + 6 = 0 \quad \therefore \lambda = -0.05 \pm 2.449i$$

$$\therefore \xi_{11} = \begin{bmatrix} 1 \\ -\frac{1}{120} + 0.4082i \end{bmatrix} \quad \text{for } \lambda_{11} = -0.05 + 2.449i$$

$$\xi_{12} = \begin{bmatrix} 1 \\ -\frac{1}{120} - 0.4082i \end{bmatrix} \quad \text{for } \lambda_{12} = -0.05 - 2.449i$$

for $x = -1$.

$$\begin{bmatrix} \ddot{x} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} -0.1 & 6 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ x \end{bmatrix}$$

$$\therefore \begin{vmatrix} -0.1 - \lambda & 6 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\therefore \lambda^2 + 0.1\lambda - 6 = 0 \quad \therefore \lambda = -0.05 \pm 2.45$$

$$\therefore \xi_{21} = \begin{bmatrix} 1 \\ \frac{5}{12} \end{bmatrix} \quad \text{for } \lambda_{21} = 2.4$$

$$\xi_{22} = \begin{bmatrix} 1 \\ -\frac{2}{5} \end{bmatrix} \quad \text{for } \lambda_{22} = -2.5$$

Draw state space.

